UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2010 question paper

for the guidance of teachers

4024 MATHEMATICS (SYLLABUS D)

4024/22 Paper 22, maximum raw mark 100

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the May/June 2010 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.

UNIVERSITY of CAMBRIDGE
International Examinations

Section A

Qu	Answers	Mark	Comments
1	**(a)** **(i)** $p = 7$, $q = 2.9(0)$	B1	
	$r = 0.25$ or $\frac{1}{4}$	B1	
	(ii) \$7.75	B1	
	(b) 0.2×980 (= 196) and 24×36 (= 864) soi	M1	Correct method for both parts
	\$80	A1	
	(c) 3.5%	B3 [8]	SC2 for answer of 23.5 or 17.5 SC1 for answer of 117.5 or 763.75 – 650 soi by 113.75 or 22.75
2	**(a)** **(i)** 110	B1	
	(ii) 10	B1ft	120 – their **(a)(i)** (provided answer > 0)
	(b) **(i)** $x + 2x - 70$ + their 10 = 180 oe or $x + 2x$ + their 110 + 70 + 120 = 540 oe	M2	Allow M2 for $2x - y = 70$ and $x + y = 170$ where $y = E\hat{D}A$ If M0, SC1 for $3x$ soi
	80	A1	NB: 80 from wrong working is M0
	(ii) 90	B1ft [6]	180 – their **(a)(ii)** – their **(b)(i)** Or 2 × their **(b)(i)** – 70 (provided answer > 0)
3	**(a)** Mercury, Mars, Venus, Earth	B1	
	(b) 3000 or 3×10^3 cao	B1	
	(c) $5.5(12) \times 10^{24}$ isw	B1	
	(d) $\frac{4}{3}\pi (6.4 \times 10^3)^3$	M1	
	1.09 to $1.1(0) \times 10^{12}$ isw	A1 [5]	
4	**(a)** $y < 12$	B1	Condone $4 < y < 12$ and $y \leqslant 12$
	y and $2x$ seen in an equality or an inequality	M1	SC1 for $y > x$
	$y > 2x$ oe	A1	
	(b) **(i)** 16	B1	
	(ii) $d = 9$ or (3, 9)	B1 [5]	

© UCLES 2010

5	**(a)** **(i)** $\begin{pmatrix}930\\1235\end{pmatrix}$ final answer After B0, column matrix with one correct or row matrix with both correct B1	B2	
	(ii) Top value – cost of fruit in week 1 Bottom value – cost of fruit in week 2	B1	
	(iii) \$21.65	B1ft	Sum of their two values divided by 100
	(b) M = $\begin{pmatrix}-6 & 0\\2 & -4\end{pmatrix}$ oe without fractions	B2	SC1 for either +4M or –4M or + or – $\begin{pmatrix}24 & 0\\-8 & 16\end{pmatrix}$
	(c) **(i)** **(a)** 7	B1	
	(b) {10, 14, 16}	B1	
	(ii) $\frac{3}{16}$	B2 [10]	SC1 for $(A \cap B =)$ {3, 6, 12} Or n$(A \cap B)$ = 3
6	**(a)** $m = \frac{1}{8}$	B1	Accept 0.12 or 0.13
	$n = 8$	B1	Accept $\frac{32}{4}$ or $\frac{8}{1}$ if correctly plotted
	(b) 5 correct central points	P2	–1 for each wrong plot –1 wrong scale –2 non-uniform scale
	Smooth curve through 5 correct central plots	C1	Lost for ruled or thick lines
	(c) **(i)** 3.5 – 3.7 ft from $y = 3$	B1	Do not accept embedded answers unless clearly justified on graph
	(ii) 2.5 – 2.7 ft from $y = 1.5$	B1	
	(d) **(i)** $t = x - 2$	B1	
	(ii) $x = \frac{5}{4}$ or 1.25 final answer	B1 [9]	Follow through their expression provided it is linear

© UCLES 2010

7	**(a)** **(i)**	184 (cm^2)	B1	
	(ii)	Tan $P\hat{S}R = \frac{8}{12}$	M1	
		$P\hat{S}R$ = 33.69 to 33.7	A1	
	(b) **(i)**	$\frac{KM}{LM} = \frac{KL}{LN}$ oe	M1	$\frac{KM}{18} = \frac{15}{10}$ oe
		27 (cm)	A1	
	(ii)	KN = 15 cm	B2	After B0, NM = 12 seen B1
	(iii)	$\frac{16}{65}$ cao	B2 [9]	B1 for unsimplified equivalents or 0.246...

© UCLES 2010

Section B

Qu	Answers	Mark	Comments
8	**(a)** $\frac{10}{x}$	B1	
	(b) $\frac{15}{x+0.5}$	B1	
	(c) their $\frac{10}{x}$ $+2+$ their $\frac{15}{x+0.5}$ $=7$ oe	B1	
	$5x\,(x+0.5) = 10x + 5 + 15x$	M1	Correct removal of the denominators x and $x + 0.5$
	$2x^2 - 9x - 2\ (= 0)$	A1	All correct – Answer given Must see at least 2 steps from previous line
	(d) For numerical $\frac{p \pm (\text{or} + \text{or} -)\sqrt{q}}{r}$		
	$p = 9$ and $r = 4$	B1	
	$q = 97$ or $\sqrt{q} = 9.848...$	B1	
	4.71	B1	SC1 for 4.7 to 4.72 and –0.2 to –0.22
	–0.21	B1	ww...max 2 marks
	(e) **(i)** 5.2(1)	B1ft	Their $x + 0.5$ (provided $x > 0$) If 2 positive values allow ft on either
	(ii) $\frac{10}{\text{their } 4.71}$ and $\frac{15}{\text{their } 4.71 + 0.5}$	M1	
	$0.75 \leqslant t \leqslant 0.8$	A1 [12]	
9	**(a)** 305° cao	B1	
	(b) $20^2 + 17^2 \pm (2) \times 20 \times 17 \cos 50°$	M1	
	$QL^2 = 20^2 + 17^2 - 2 \times 20 \times 17 \cos 50$	M1	
	15.87 – 15.9	A2	After A0, 251.9, 252 SC1
	(c) **(i)** $\frac{\sin P\hat{L}Q}{20} = \frac{\sin 50}{\text{their } 15.9}$	M1	
	$\sin P\hat{L}Q = \frac{20 \sin 50}{\text{their } 15.9}$ (= 0.9653)	M1	Dep on first M1
	$P\hat{L}Q$ = 74.48 to 74.9	A1ft	ww 2 marks
	(ii) (0)19.48 to (0)20	B1ft	Their **(c)(i)** – 55
	(d) **(i)** 2130 or 9 30pm	B1	Not 09 30 (pm)
	(ii) $\sin 50 = \frac{x}{17}$ or $\sin Q = \frac{x}{QL}$	M1	
	x = 12.9 to 13.1 (km)	A1 [12]	

© UCLES 2010

10	**(a)** $n = 4$ 22, 20, 42 $n = 5$ 26, 30, 56	B2	After B0, 4 correct values SC1
	(b) **(i)** $4n + 6$	B1	Accept $2(2n + 3)$ or $4 \times n + 6$
	(ii) $n^2 + n$	B1	Accept $n(n + 1)$ or $n \times n + n$
	(c) $n^2 + 5n + 6$ $(n + 2)(n + 3)$	M1 A1	Adds their expressions for **(b)(i)** and **(b)(ii)** Factorises – answer given NB: Alternative complete methods can score M1A1
	(d) 156	B1	
	(e) **(i)** $((k + 2)(k + 3) = 306)$ $k^2 + 5k + 6 = 306$ $k^2 + 5k - 300 = 0$	M1 A1	
	(ii) 15 –20	B1 B1	SC1 for –15 and 20
	(iii) 66	B1ft [12]	Their positive integer k substituted into their **(b)(i)**
11	**(a)** **(i)** Correct scales and Correct widths (2, 2, 5, 5, 10) Correct heights (6, 9, 8.4, 5.6, 4)	SW1 H2	 3 or 4 correct heights H1
	(ii) 21 or 20	B1	
	(iii) $\frac{5}{7}$ cao	B1	
	(iv) $\frac{132}{870}, \frac{22k}{145k}$ or 0.15(0) to 0.152	B2	SC1 for $\frac{132}{900}, \frac{11k}{75k}$ or 0.147 or $\frac{12 \times 11}{30 \times 29}$ or $\frac{132}{870}$ seen
	(b) **(i)** $\frac{7}{60}$ cao	B1	
	(ii) 60	B2	After B0, 35% = 21 seen SC1
	(iii) 8	B2 [12]	SC1 for either 15,21 and 7 seen or 48° or 13 ⅓ % seen

© UCLES 2010

12	**(a)** **(i)** 15	P2	After P0, $\sqrt{9^2+12^2}$ P1
	(ii) 678 – 679 (cm^2)	S2	After S0, $\pi \times 9 \times$ their $15 + \pi \times 9^2$ S1
	(iii) 1017 – 1020 (cm^3)	V2	After V0, $\frac{1}{3} \times \pi \times 9^2 \times 12$ V1
	(b) **(i)** 4 cm	B1	
	(ii) 10 cm	B1	
	(iii) 18.8 – 18.9 (cm)	C2	After C0, $\pi \times 3 \times 2$ C1
	(iv) 979 – 983 (cm^3)	W2	After W0, $\frac{26}{27}$ × their 1018 or
		[12]	their $1018 - \frac{1}{3}\pi 3^2 \times$ their 4 W1

© UCLES 2010